

Body condition mediates carry-over effects of a deteriorating overwintering environment on reproduction in a declining shorebird

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Abstract

Life cycles consist of linked stages, such as reproduction and overwintering. Carry-over effects (COEs) via body condition can postpone the impacts of environmental change to later stages and seasons. This complicates identifying drivers of population change, as these occur in other seasons, and for migratory species also in other places, than their impacts. Furthermore, COEs are rarely studied at a landscape scale or rigorously quantified using path analysis, so their role in population change remains poorly understood. We investigated COEs in a declining shorebird (*Haematopus ostralegus*) by linking poor environmental conditions at communal wintering grounds to low breeding success across habitats. We advanced beyond the traditional focus on mass, applying a multifaceted approach by measuring multiple physiological aspects of body condition in 1574 individuals. We followed them across their annual life cycle using a 20-year nationwide citizen science effort and applied an analytical framework integrating structural equation modelling and capture-recapture analysis to quantify the COE and its mechanism. Winter body condition predicted reproductive success, indicating a COE. Offspring from parents with high winter body condition survived longer the next summer, but not due to earlier breeding. Winter body condition was higher where grassland cover was greater, while individuals with different diet specialisations responded differently to temperature and food availability. Grassland decline, estimated at 40% over recent decades, can explain a 13.5% reduction in offspring, supporting our hypothesis that reproductive decline throughout the Netherlands is partly due a COE from a deteriorating winter environment. Causes thus do not necessarily lie in the breeding environment. Though grasslands are often seen as marginal habitat, our findings suggest they are vital for maintaining these shorebirds. We recommend preserving and restoring grasslands around major winter roosts. Our multifaceted, landscape-wide approach offers a template for overcoming challenges in studying COEs and identifying drivers of population decline.

Introduction

One of the central goals in population biology, ecology and conservation is to identify the drivers for fluctuations in population abundance (Betini et al. 2013, Brown et al. 2017). However, it can be challenging to reach this goal because many organisms (e.g. birds, fish, mammals, butterflies; Calvert et al., 2009) are

organized into annual cycles that involve different stages (e.g. reproduction, migration, overwintering, metamorphosis/moult) that are often temporally and spatially separated (Marra et al. 2015, Briedis et al. 2018). Although these stages are distinct, they are linked physiologically and ecologically, as preceding stages may have profound consequences for the following stages in the annual cycle (Harrison et al. 2011). Processes taking place in one stage (season) that result in individuals making the transition between seasons in different states (e.g. body condition levels) and consequently affecting their performance in subsequent seasons are called carry-over effects (Norris and Marra 2007, Harrison et al. 2011). Through carry-over effects (COEs), environmental conditions in one season can delay impacts until a later season, complicating our ability to pinpoint the drivers of population change, particularly as these drivers may occur in separate seasons—or, for migratory species, in different geographic locations. This complexity underscores the relevance of COEs to environmental decision-making.

Although there may be theoretically clear biological links between the different stages that animals experience throughout their annual cycle, our understanding of the ecology and physiology of wild animals is still heavily biased to single-season studies (e.g. breeding season) (Marra et al. 2015). This bias likely exists due to the logistical challenge of following individuals across seasons and the analytical challenge in identifying the complex interactions across seasons between individuals and their environment. However, to improve conservation and management decisions, advancing our understanding of how natural and anthropogenic changes affect both individual and population-level responses across seasons is vital (Marra et al. 2015). For example, single-season studies may misclassify the mechanisms underlying population dynamics, which can generate incorrect projections of population numbers under future anthropogenic change, misguiding conservation efforts.

COEs in migratory species are particularly challenging because migrants not only live in different areas in different seasons, but individuals from the same breeding (or overwintering) area may overwinter (or breed) in different areas (Norris et al. 2004, Gunnarsson et al. 2005). For example, the COE on reproduction could be mediated by animal's body condition in different wintering locations, which in turn is affected by spatially varying environmental conditions. Variation in reproductive success among breeding areas can then only be understood by investigating (i) the environmental conditions beyond the breeding area (e.g. on the winter grounds) and (ii) the migratory connectivity between wintering and breeding areas. As another example, if individuals from different breeding populations communally overwinter in the same area, conditions on the wintering ground may explain temporal variation in reproduction across many populations. This complexity emphasizes the need to study COEs on a large spatial scale, critical for ecological applications and management, but landscape-scale year-round tracking poses major logistical challenges.

Several studies investigated COEs by linking environmental conditions (e.g. winter climate, food availability) in one season directly to individuals' performances (reproduction, survival) in the subsequent season (Harrison et al. 2011), without considering which state variable of the individual mediates the COE. However, erroneous conclusions may be drawn if one does not explicitly consider the mediating mechanistic role of state variables such as an individual's body condition. For example, Olivier et al. (2005) found that snow petrels (*Pagodroma nivea*) showed higher reproductive success after winters with greater ice cover. This may appear as a COE where the environment in the non-breeding season affects the reproduction in the next season. However, the authors suggest that sea ice also promotes overwinter survival of krill, and thereby ensures a larger supply of food during the breeding season, consequently resulting in increased reproductive success. Environmental conditions in one season may thus be proxies of environmental conditions in subsequent seasons, but if the causal environmental drivers of organismal performance are from the same season, they are not COEs. Furthermore, the predictive value of proxies from other seasons could diminish under global change. Thus, identifying individual state variables (e.g., body condition, arrival date) that mediate COEs of environmental conditions on subsequent performance is essential for improving ecological decision support.

Whereas most studies look at body mass as the key individual state variable that mediates COEs across seasons, two individuals could be of similar body mass, and thus appear superficially to be in similar con-

dition, but in practice may still have different abilities to deal with environmental challenges (Harrison et al. 2011). In fact, in the past decade it has become clear that body condition is also related to aspects other than body mass, such as health status, which can even be even more appropriate predictors of fitness (Barry and Wilder 2013, Nie et al. 2014). In addition, several studies have suggested that hormones and the immune system may be involved in COEs (McNamara and Houston 2008, Crossin et al. 2012, Hegemann et al. 2015). Therefore, to identify the mechanism of a COE, it is essential to measure various condition variables (e.g. nutritional, hormonal, immune state) in one season, link it to environmental drivers in that season, and relate it to individual performance in the next season, which is rarely done.

To accommodate all the above biological complexities and interrelationships underlying COEs, one needs a robust analytical framework that enables their analysis in an efficient and integrative way (Marra et al. 2015, Souchay et al. 2018). Specifically, the survival of wildlife is typically studied by recapturing or resighting marked individuals, which is analysed using capture-mark-recapture (CR) analysis. But CR models struggle with time-varying individual covariates such as body condition (Cooch and White 2019). Furthermore, body condition is a multidimensional concept, and recent studies have suggested that latent variables can be efficient in operationalizing the concept of body condition from multiple measurable body condition traits in the field (Frauendorf et al. 2021). Finally, COEs of conditions in one season to another must be mediated by a state variable such as body condition, and path analysis provides a logical tool to identify the relative importance of various mediating pathways. Fortunately, methodological advances have been made where CR can be combined with structural equation modelling (SEM; a generalization of path analysis that inter alia also allows for latent variables) to integrate the above processes into a single integrative Bayesian framework (Cubaynes et al. 2012). However, despite CR-SEM's suitability for the analysis of COEs (which by definition involves indirect pathways and rampant imperfect detection due to the challenges of following many individuals across seasons), it has not been applied to this context yet.

The Eurasian oystercatcher (*Haematopus ostralegus*) experienced a strong population decline in recent decades in many parts of its range (van de Pol et al. 2014) and is listed as a near-threatened species (BirdLife International 2023). The primary bottleneck for this decline –from a demographic perspective– is reduced reproductive success (Hulscher and Verhulst 2003, Roodbergen et al. 2012, Allen et al. 2022). The Dutch oystercatcher population is partially migratory: in winter they aggregate in the Wadden Sea and Delta estuary, while in summer they spread out over the country to breed in a variety of breeding habitats (coastal saltmarshes, inland agricultural areas, urban roof nesting). Reproductive success is low in most breeding areas, however, it seems unlikely that one or a few environmental drivers in the breeding season can explain the nationwide pattern, as breeding habitats vary widely in their food sources (benthic vs. terrestrial), predator communities (mammal vs. avian dominated; Frauendorf et al., 2022), human influences (urban vs. nature reserves) and weather impacts (flooding of coastal nests. vs. overheating on urban roofs). However, as birds from different breeding areas share overwintering grounds, we hypothesize that there could also be one or a few common drivers in their overwintering area leading to reduced conditions in winter that carry over to the breeding season, to cause the overall low reproductive success that drives the population decline of this species insights directly relevant for informing conservation and management strategies.

In this study, we advance the existing knowledge of COEs by (i) clarifying the role of body condition in mediating COEs on reproduction, (ii) identifying which winter environmental effects carry over on the reproductive success, and (iii) quantifying by how much COEs of deteriorating winter environmental conditions have reduced reproductive success in recent decades, such that we can assess the relative importance of COEs for population dynamics in this declining species. We took a landscape-scale approach by catching 1574 individuals across the Netherlands to measure multiple morphological and physiological aspects of body condition, while a 20-year nationwide citizen science program followed birds throughout their annual life-cycle. The relative importance of the pathways via which environmental change in winter could affect summer reproduction were quantified using an integrative Bayesian CR-SEM framework (Cubaynes et al. 2012). Specifically, we (1) operationalized and quantified body condition of wintering oystercatchers using multiple morphological and physiological variables to (2a) test if winter body condition directly carries over to the reproductive performance in the subsequent breeding season, (2b) but also indirectly through timing

of egg-laying (because reproductive success strongly depends on breeding phenology and may therefore be a reproductive trait that mediates how body condition affects reproductive output). Next, (3) we identified to what extent environmental conditions (food availability and weather) affected winter body conditions. Finally, (4) we use literature-based estimates of historical rates of environmental change of winter conditions and combine it with steps 1-3 to quantify by how much reproductive success may have decreased over the past decades due to COEs of a deteriorating winter environment (for an overview see Figure 1a).

Method

Data collection & preparation

Study area

We caught and sampled 1574 oystercatchers in the two main estuaries of the Netherlands. Birds were caught with cannon nets or mist nets during 36 catching events around the Dutch Wadden Sea and Delta area in two periods (winters of 2000-2003: 837 birds; winters of 2016-2018: 737 birds; Figure 2; Appendix S1: Table S1).

Body condition variables

We considered two physiological and one morphological body condition variable: body mass corrected for size (as a proxy for fat storage), haematocrit and buffy coat. Buffy coat, the fraction of white blood cells, is elevated if the body needs to fight against infections (Campbell et al. 2008), and tends to be elevated in individuals in adverse conditions (Ullman-Culler 1999, Beechler et al. 2012). Haematocrit, the proportion of red blood cells, facilitates oxygen transportation. Buffy coat and haematocrit have been shown to predict survival in Eurasian oystercatcher (Verhulst et al. 2004).

Blood samples (approximately 0.35 ml per bird) were taken from the brachial vein. Two capillary tubes of approximately 65 μ L were taken from each bird and centrifuged 10 minutes at 9503g force between 2 to 6 hours (mean $\pm$ se) = 3.6 $\pm$ 0.96 hrs) after blood extraction. Since this time delay did not significantly affect buffy coat and haematocrit, it was not included as a variable in the analysis. Buffy coat and haematocrit were measured by taking standardized pictures of the blood collection tube in a specific holder constructed for this purpose (Appendix S1: Figure S1), and then measuring the length of the portion of the tube that contained red blood cells, white blood cells or plasma in pixels with the program Paint.NET. The haematocrit and buffy coat were calculated by taking the proportion of the total length of red blood cells and white blood cells, respectively (total=red blood cells+ white blood cells + plasma). The reason for using pictures and measuring pixels instead of measuring proportions directly in the field was to minimize measuring imprecision due to more accurate measurements compared to a calliper. Repeatability between the two samples per individual was high (on average 0.92; Appendix S1: Table S2), therefore, we calculated the average of both for each haematocrit and buffy coat per individual. The repeatability was calculated by estimating the intra-class correlation coefficients (ICC) (and its 95% confidence interval) and using the variance components from a one-way ANOVA with the R-package ICC (Wolak et al. 2012). Since haematocrit and buffy coat values ranged from 0 to 1, they were logit transformed before analysis (Warton and Hui 2011).

Body mass (g) and various measures of structural size (length of tarsus to toe, wing length, head length, bill tip height) were measured to the nearest mm following standard techniques described in Durell et al. (1993). Handling time, the time between capture and measuring (a confounding variable; Figure 1a-b), was recorded to correct for time-dependent mass loss as well as possible effects on other physiological measures (Verhulst et al. 2004, C r le et al. 2012).

Individual characteristics

Sex was determined using molecular techniques (DNA) from blood stored in cell lysis buffer at room temperature. Across all sampling years, 37% of the birds were not sexed using DNA because too little or no blood was taken. Sampling during winters 2001-2002 and 2002-2003 had less than 5% sexed using DNA, whereas in the remaining years, >80% of the birds were sexed using DNA (Appendix S1: Table S3). Birds that were

not sexed using DNA, were sexed using a classification tree function (Hothorn et al. 2006) from the partykit R-package (Hothorn and Zeileis 2015). Previous studies have shown relationships between the sex and biometric measures (Durell et al. 1993, Zwarts et al. 1996a). Because sexual dimorphism varied substantially in time and space (van de Pol et al. 2009a), we only included birds caught in winter for the classification tree and also added the sampling year and catching location as additional variables. 75% (n=1061) and 25% (n=281) of observations has been used as training and test dataset, respectively. Most important biometric variables were bill length, bill tip height and bill shape (Appendix S1: Figure S2), which resulted in 90% accuracy with a 95% confidence interval of 86% and 94% of correct classification (Appendix S1: Figure S3).

Birds were aged based on their plumage, bill and leg characteristics (Cramp and Simmons 1983) and the age was classified in 1st, 2nd, 3rd calendar year and adults (>3rd calendar year). It is not possible to accurately age adult birds. Furthermore, bill tip height (measured 3 mm from the bill tip using a calliper) was used as a proxy for the type of individuals' diet specialization (individual characteristics; Figure 1a-b), ranging from worm specialists (indicated by a pointed bill and a low bill tip height) to shellfish specialists (indicated by a blunt bill and high bill tip height; van de Pol et al., 2009a).

Marking, resighting and dead recoveries

All captured birds were ringed with a unique combination of inscribed colour rings and a metal ring with a unique number. Resightings were predominantly recorded by citizen scientists across the Netherlands in an online portal (for more details on this nationwide program, see Allen et al., 2019a). The metal ring data and subsequent dead recoveries are managed by the Dutch Centre for Avian Migration and Demography. Resightings and recoveries were collated over the period 2000 to 2019, with in total 3616 resightings and 49 recoveries of the 1574 caught birds. Note that although we are mainly interested in estimating survival in the period directly following capture, which involves a few years, later resightings and recoveries improve survival estimation when detection is incomplete. Colour-ring wear and loss have negligible effect on the resighting and survival estimates (Allen et al. 2019b).

Winter environmental variables

We considered environmental variables reflecting food availability (both intertidal and terrestrial), conspecific density and weather conditions during winter. Intertidal food availability was calculated by multiplying food abundance (cockles and mussels in gram ash-free dry mass per m²; Appendix S1: Figure S4a) by the exposure time (averaged over the winter during which birds were caught; Appendix S1: Figure S4b) of the tidal flats.

Cockles (*Cerastoderma edule*) and mussels (*Mytilus edulis*) are among the principal food sources of oystercatcher in winter (Hulscher 1996), and we used a 7 km-radius area around the catching location; this radius size was chosen based on movement distances of GPS-tracked oystercatcher around several high tide roosts in the Netherlands (Bakker et al. 2021). For details on calculations of cockle and mussel abundance and exposure time see Appendix S1: Text S1. Although oystercatchers predominantly feed on intertidal flats, especially worm specialists may also feed on inland fields during high tides (Heppleston 1971, Goss-Custard and Durell 1983, Durell et al. 2001, van der Kolk et al. 2019). Therefore, we calculated the proportion of available foraging areas (grasslands) of the terrestrial area within each 7 km buffer, using data on land use from CBS Statistics Netherlands (Figure 4d; Appendix S1: Figure S5).

Conspecific density in the winter of capture was calculated from monthly high-tide roost counts organized by Sovon Dutch Centre for Field Ornithology (Hornman et al. 2012). We used the counts in November and January since those were most complete for the catching years. Within a buffer of 7 km-radius around each catching location, we averaged the numbers of counted oystercatchers for these months, and weighted them for the counted area (not the whole radius area was always counted; Appendix S1: Figure S5), resulting in oystercatcher numbers per km² (Appendix S1: Figure S4c).

Precipitation and temperature may influence food availability and directly affect the energy required for homeostasis of adult birds (Goss-Custard 1984, Kersten and Piersma 1987, Camphuysen et al. 1996). In addition, winter severity has previously been described as an important environmental factor impacting the

reproduction and survival of oystercatchers (van de Pol et al. 2010, Schwemmer et al. 2014). The windchill index was also considered to investigate the combined effects of low winter temperatures and high wind speeds given the additional energy requirements (Wiersma and Piersma 1994, Smith et al. 2012).

Weather data (KNMI 2020) were extracted for the nearest weather station from the catching location (mean (sd)=17.4km (12.1), n=36; Appendix S1: Table S4; Figure 2). For each catching location, we calculated the weather conditions over a period of 1 or 2 months before catches, since we expect that the period directly preceding the catching event rather than an average of the whole winter season may affect the condition of birds. Weather variables from 1-month and 2-month prior catching showed high correlation (ranging from 0.74 for precipitation anomaly to 0.89 for winter severity; Appendix S1: Table S5). For further details on the calculation of weather variables, see Appendix S1: Text S2.

Reproductive parameters

We attempted to locate the breeding ground in the subsequent season for as many individuals as possible that were caught in the winters of period 2. Out of 737 caught in winter in this period we found the breeding territories of 32 birds, scattered across the country. Breeding success of located individuals was monitored using Reconyx camera traps placed approximately 2m from the nest to minimise disturbance at the nest. Every 7-10 days, batteries and SD cards were replaced. The number of eggs, hatching date and number of hatched chicks was recorded. The timing of egg laying (henceforth lay date) was back-calculating from the hatch date, and for nest with unknown hatch date it was calculated from the ratio of egg mass to egg volume (eggs lose weight during incubation; Jager et al., 2000; Strijkstra, 1986), giving an lay date accuracy of $\pm$ 1-2 days. We only included nests in the analysis for which we could be confident that it was the first clutch of the year. After hatching, the precocial chicks were checked every 2-3 days to determine hatchling survival, until they either fledgling (around 4-5 weeks) or there was no sign anymore of the hatchling being alive (e.g. no alarming parents around).

Daily clutch survival was defined as the probability that a clutch survived until the next day, which was calculated using the Mayfield method that takes into account the number of observation days (Aebischer 1999). For hatchling survival, we calculated the maximum number of days the chicks were alive and assumed that fledging is reached as soon as the chick can fly (typically between 30-35 days).

Data analysis

Analysis consisted of four steps: (1) operationalize and quantify body condition, (2a) test if winter body condition carries over to the reproductive performance in the subsequent breeding season (directly) (2b) but also indirectly through timing of egg-laying, (3) identify to what extent environmental conditions affect winter body condition, and (4) use literature-based estimates of historical rates of environmental change of winter conditions and combine it with steps 1-3 to quantify how much reproductive success may have decreased due to COEs of a deteriorating winter environment (for an overview of step 1-3 see Figure 1a).

All analytical steps used a Bayesian approach for estimation and inference. Bayesian models were constructed using R-package nimble (de Valpine et al. 2017). Bayesian R^2 were estimated using equations described in Gelman et al. (2019). Next to documenting the mean and 95% credible interval of the posterior distribution for all model results, we also documented the probability of direction (p_d) that can be interpreted as the Bayesian equivalent of the complement of the frequentist p-value (Makowski et al. 2019). p_d is the posterior probability that a derived parameter is positive or negative, whichever is the most probable. This means that the p_d -value equals the proportion of the posterior density that has the same sign as the median of the posterior distribution. We describe the p_d -values between 0.95 and 0.97 as providing ‘weak evidence’, a p_d -value between 0.97 and 0.99 as ‘moderately strong evidence’ and a value larger than 0.99 as ‘strong evidence’ (Muff et al. 2022).

Step 1: Operationalize body condition

We fitted a multi-site capture-recapture model combined with a structural equation model (CR-SEM; Curbaynes et al., 2012) to quantify body condition (Figure 1b,c). The variable ‘body condition’ was modelled as a

composite variable, summarizing the multivariate concept of body condition, representing the collective effect of multiple observed variables (Grace and Bollen 2008, Frauendorf et al. 2021): body mass, haematocrit and buffy coat. In addition, confounding variables (handling time, catching day) and individual characteristics (sex, age, bill-tip height) were included that may affect the condition variables, and that we want to correct for (Figure 1b). By considering effects of handling time on body mass, we accounted for time-dependent mass loss. Larger birds are heavier, and therefore body mass was corrected for size using three body size metrics (head, wing and tarsus length). Thus, we henceforth focus on size-corrected mass as a proxy of fat content.

All continuous variables (mass, haematocrit, buffy coat, tarsus length, head length, wing length, bill-tip height, catch day, handling time) were standardized to z-scores. Age and sex were included as categorical variables. Age was divided in three categories: 1-year-old (n=244), 2-year-old (n=153) and ≥ 3 -year-old (n=1197) birds. We grouped 3-year-old and ≥ 3 -year-old birds, because we only had sixteen 3-year-old birds in the dataset. We expected especially the 1- & 2-year old birds to have different body condition from adults (≥ 3 year) rather than the 3-year-old birds, as no survival differences were found between 3-year-olds and adults (Allen et al. 2022). A squared term for handling time was included (following Verhulst et al., 2004).

Since there may also be a non-linear relationship of mass, haematocrit and buffy coat on the individual body condition (e.g. birds with very low or very high mass being in lower condition than birds with average mass), we compared the Watanabe-Akaike information criterion (WAIC) for a model with only linear terms (for mass, haematocrit and buffy coat) with a model with an additional squared term for each condition variable. The WAIC of the model with the linear term only was lower (Appendix S1: Table S6) and visualization of the relationships confirmed that relationships were approximately linear (Appendix S1: Figure S6).

To further operationalize body condition, we assumed that it must be a predictor of overwinter survival. This implies that the composite variable body condition can be viewed as an independent variable that explains variation in the response variable survival. *Therefore, we next* describe the capture-recapture model used to estimate survival and how it was combined with the body condition SEM (from the previous section) into a CR-SEM. Combining structural equation modelling with capture-recapture models allows the integration of imperfect detection with the advantages of SEM to test hypotheses within a complex structural framework (Cubaynes et al. 2012).

We used a seasonal multi-site live dead recovery model with a time-varying individual covariate and a site-specific random time-effect (Figure 1c). We used eight geographical areas, hereafter sites, to estimate survival, resighting probability and seasonal transitions (migration) among areas. The eight geographical sites were chosen based on previous knowledge about spatial heterogeneity in survival, migration and resighting probabilities. We divided the inland sites into north and south due to the expected variation in transition probabilities, since oystercatchers breeding inland in the southern part of the Netherlands were more likely to migrate to the Delta and further south, while oystercatchers breeding inland in the northern part of the Netherlands were more likely to migrate to the Wadden Sea (Allen et al. 2019a). The coastal areas of the Wadden Sea and the Delta estuary were separated due to expected variation in survival resulting from past activities related to human activities like shellfisheries and habitat loss (van de Pol et al. 2014). The Dutch Wadden sea estuary was subdivided further into five sites, reflecting tidal basins that often encompass the home range of birds in a given season. The site (geographical location) of an individual was coded according to one of the eight geographical locations (Figure 2): D=Delta, P=Inland south, N=Inland north, B=Balgzand/Texel, V=Texel/Vlieland, T=Terschelling/Ameland, S=Schiermonnikoog and R=Rottum. The ninth site 'X' (abroad) captured all observations outside the Netherlands (Allen et al. 2019a). The dead recovery was coded as a separate state in the encounter history, but did not contain information about where the bird was found (0/1). The sampling period was 20 years (2000-2019), but the time interval in the model was half a year to account for the seasonal structure of our model, thus providing a total of 40 time intervals. In addition, we added an age effect to estimate survival for each age class: juveniles (1-year-old individuals), sub-adults (2 -and 3-year old individuals) and adults (≥ 3 -year-old individuals) (Figure 1c).

Finally, we used the body condition from the SEM as an individual covariate in the CR model (Figure 1c).

This approach allowed us to assess the influence of ‘body condition’ of individual i in winter t , on survival to $t+1$, rather than treating condition as a static trait being constant across years, i.e. we treated body condition as a time-varying individual covariate that was only related to survival in the first time interval after capture. For details on prior choice, age-structure, additional confounding variables, and technical details of the Bayesian analysis, see Appendix S1: Text S3.

Step 2: Relate body condition to reproductive variables

To determine if a COE exists, we conducted path analysis with the R-package *brms* (Bürkner 2017, 2018) to investigate the effect of body condition on reproductive performance. We considered a direct effect on reproductive performance as well as an indirect pathway mediated via the timing of egg laying (Figure 1e). Since avian reproductive success is highly affected by breeding phenology (i.e. lay date; Low et al., 2015; Reséndiz-Infante et al., 2020), we may expect that COEs of body condition could be mediated by the timing of egg laying (Jean-Gagnon et al. 2018, Montreuil-Spencer et al. 2019).

Since we only had a small subset of birds (max. $n=32$) for which we had information on both their condition in winter and on reproductive success in the subsequent breeding season, we ran the path analysis in a separate model rather than in the same model as the CR-SEM (Figure 1a). Both sexes contribute equally to incubation, clutch defence and offspring feeding in oystercatchers (Ens et al. 1996). Therefore, we expect that the body condition of either sex could be a state variable mediating COEs and therefore did not differentiate between males and females in the carry-over effect analysis.

We ran two separate path analyses using different measures of reproductive performance, one with daily clutch survival probability ($n=32$), and one with hatchling survival ($n=25$) as the response variable. For the path analysis with ‘daily clutch survival’ as a response variable a beta distribution was used, for ‘hatchling survival’ (in days) we used a Poisson distribution, while a Gaussian distribution was used for the response variable ‘lay date’. We standardized lay date separately for mainland and island breeding areas, because inland breeding birds started their clutch on average 10 days earlier (Appendix S1: Figure S7). Birds breeding in cropland showed significantly lower hatchling survival (Appendix S1: Figure S8b), whereas body condition did not differ significantly across habitat types (Appendix S1: Figure S8a). This may influence the relationship of body condition on hatchling survival (Appendix S1: Figure S8c) and we therefore included a factor ‘cropland’ (no/yes) to the path analysis on hatchling survival. We did not include any habitat variable in the clutch survival path analysis because there was no substantial effect of habitat type (Appendix S1: Figure S9). We used two MCMC chains of 15 000 iterations, a burn-in period of 4000 iterations and a thinning of 1 that resulted in acceptable mixing and convergence (Appendix S1: Figure S10-S11; Appendix S1: Table S7-S8).

Step 3: Relate winter environmental conditions to body condition

To assess the impact of winter environmental conditions on body condition, we related winter environmental variables to body conditions (Appendix S1: Fig 1a). Variable reduction of the highly correlated weather variables was implemented within our SEM (Appendix S1: Figure S12). We combined temperature-related variables (average temperature, wind chill index and winter severity) into one latent temperature variable and precipitation related variables (precipitation sum, precipitation anomaly) into another latent precipitation variable (Figure 1d). We investigated linear as well as non-linear effects of all five environmental variables (temperature, precipitation, cockle, and mussel availability as well as grassland proportion), but the model fit with only linear terms was better supported in all cases (lower WAIC; Appendix S1: Table S9). Finally, we added a random effect for capture (Appendix S1: Figure 1d; capture ID in Appendix S1: Table S1) because individuals caught at the same site and year may show similar patterns, violating assumptions about independence. Because correlation between the food condition variables was low (Appendix S1: Figure S12), we investigated the effect of cockles, mussels and grasslands on condition as separate predictors.

Previous studies have shown that individuals with different diet specialization may respond differently to weather variability (van der Kolk et al. 2019). Furthermore, a bird’s diet specialization is expected to moderate how they are affected by specific food stocks (shellfish specialists mainly by cockles and mussels, worm specialists more by grassland). Therefore, we also included an interaction term between diet specialization

and environmental variables.

Since we had no prior knowledge on which time period (1 month or 2 months) before the catching event may be crucial in terms of how weather condition influence an individual's body condition, we initially conducted analyses separately by using the weather conditions one month and two months before the catching event. Based on the WAIC of both models, environmental variables collected 1-month prior to catching fitted the data better and were used in the remainder of the analysis (Appendix S1: Table S9).

Results

Operationalizing and quantifying winter body condition

Size-corrected mass (a proxy of fat content) was negatively associated with winter body condition (Figure 3a; $\gamma_1 = -1.55[-2.61; -0.63]$, whereas haematocrit and buffy coat were positively associated with condition ($\gamma_2 = 0.61[0.09; 1.15]$, $\gamma_3 = 0.61[0.00; 1.33]$, respectively; Figure 3d-f). These relationships were consistent in their direction across the two study sub-periods (2000-2003; 2016-2018; Appendix S1: Figure S13). Our model enforces a positive relationship between body condition and higher survival probability. Notwithstanding, the effect of body condition was very strong ($\beta_1 = 0.74[0.47; 1.01]$), with the predicted survival increasing from 0.2 to virtually 1 when comparing birds with the lowest and highest body condition in our study (Figure 3g). Furthermore, body condition explained a substantial amount (25%) of the variation in survival across sites, years and age classes.

As expected, body mass was positively correlated with all three metrics of body size (Figure 3a). Size-corrected mass, haematocrit and buffy coat were further corrected for the effect of multiple potential confounding variables (Figure 3a-c). Furthermore, individuals who had to wait longer before being handled (after their capture) had a lower size-corrected mass and buffy coat, but higher haematocrit (Figure 3a-c). In addition, birds caught later in the winter season show lower size-corrected mass and buffy coat, but higher haematocrit. Older birds were heavier but had lower haematocrit and buffy coat, while males had a slightly higher haematocrit and buffy coat than females. Finally, shellfish specialists (i.e. bird with a high bill tip height), had a higher size-corrected mass, buffy coat and a lower haematocrit than worm specialists.

Carry-over effects of winter body condition on reproductive success

There was moderately strong evidence of an effect of both body condition and lay date on reproductive success, as regression analysis revealed a positive association between individual winter body condition and lay date on both the clutch and hatchling survival (Figure 4a,c). However, the positive effect of body condition on reproductive performance did not appear strongly mediated by earlier laying, as (i) there was no evidence for an effect of winter body condition on lay date (Figure 4b; $p_d < 0.94$), and (ii) the direct pathway of winter body condition on clutch and hatchling survival was much stronger (0.34 and 0.13, respectively) than the indirect pathway via lay date (-0.04 for both clutch and hatchling survival; Figure 4). In addition, birds breeding on crop fields showed lower hatchling survival and later lay date (Appendix S1: Table S8).

Environmental drivers of body condition in winter

We found moderately strong evidence for an effect of grassland proportion on body condition in winter (Figure 5f; $0.24[0.02; 0.45]$; $p_d = 0.98$). We found no evidence for an overall effect of conspecific density, temperature, precipitation, cockle or mussel availability on the winter body condition as the probability of direction (p_d) is smaller than 0.95 for all these variables (Figure 5a-e). However, we did find further support for some environmental variables affecting the body condition of specific type of individuals (Figure 6b). Specifically, our model shows moderately strong evidence for worm specialists being positively affected by higher temperatures ($0.28[0.02; 0.54]$; $p_d = 0.98$; but not for shellfish specialists $-0.02[-0.32; 0.28]$; $p_d = 0.90$). Generalists lie in-between the two specialist feeders with a mean temperature effect in the positive range ($0.17[-0.06; 0.40]$; $p_d = 0.92$). Furthermore, the body condition of shellfish specialists benefited more from higher cockle availability ($0.09[-0.16; 0.36]$; $p_d = 0.75$) than did worm specialists ($-0.13[-0.36; 0.11]$; $p_d = 0.88$), however being not strictly positive or negative, respectively (Figure 6b). Again, the generalists could be found in the middle of the two specialists ($0.00[-0.19; 0.20]$; $p_d = 0.51$). Individuals with different diet specializations

also show different mean body condition with worm specialists having a lower body condition ($-0.36[-0.54;-0.17]$) compared to shellfish specialists ($0.12[-0.02;0.27]$; Figure 6a). Generalists show a body condition lying in-between the two specialist feeders ($0.01[-0.14;0.16]$).

Carry-over effects of changes in winter environment on reproductive output

To facilitate further biological interpretation, we calculated the total COE of three winter environmental variables on hatchling survival mediated by body condition. We only considered environmental variables that had an overall effect on body condition (grassland), as well as those that affected the body condition differently for different diet specializations (temperature and cockle availability, Figure 6b). We found that a 1°C increase of winter temperature increased the hatchling survival (which averaged 18.2 days) by 0.4 days ranging from -0.6 to 1.5 days (95% confidence interval), which corresponds to an on average 2.4% increase in hatchling survival (Figure 7, Appendix S1: Table S11). Similarly, an increase of grassland proportion around the wintering ground by 0.13 (being equivalent to 1 SD) results to an average increase of hatchling survival by 0.8 days $[-0.6;2.3]$, which corresponds to an increase of 4.5% (Figure 7, Appendix S1: Table S11). An increase of cockle availability by 10 g AFDM/m² (being equivalent to 1 SD) results in a 1.7% increase in hatchling survival (on average 0.3 days, Figure 7, Appendix S1: Table S11).

Contribution of COEs to historical declines in reproduction

The size of permanent grasslands decreased by almost 40% across the Netherlands since 1980s (CBS et al. 2021), which implies that over this period, assuming a linear relationship, this decline in grassland is expected to have led to a reduction of 13.5% in hatching success (Fig 7; 13% change in grassland * 3 results in 40% change in grassland size; 4.5% change in chick survival * 3 results in 13.5% change in chick survival). Winter temperatures became milder in the last decades, and thus cannot explain low(ered) reproductive success in recent decades, as warmer winters imply a positive carry over effect on hatching success (Figure 7). Similarly, there is no clear evidence of a consistent change in the highly variable cockle stocks in the period 1980-2020 (Beukema & Dekker, 2006; Troost et al., 2021; Figure 7), and thus they are unlikely to explain overall reductions in reproduction over time observed in oystercatcher populations in the Netherlands.

Discussion

Carry-over effects, by delaying the impacts of environmental change to later seasons, introduce significant challenges for identifying drivers of population change, especially when those drivers occur in different seasons or regions than their effects. This challenge is amplified in migratory species, necessitating longitudinal studies across extensive migratory networks and careful identification of the mediating state variables of COEs. Our study addressed these obstacles by implementing a multifaceted approach, combining landscape-wide catching efforts and physiological assessments with long-term citizen science data and an integrative analytical framework. This approach enabled us to quantify the significance of COEs for a near-threatened species and to address a critical hypothesis: that the nationwide decline in reproductive success is not rooted in disparate breeding grounds but in the shared overwintering areas, underscoring the need for policy and management solutions at a landscape scale.

Our study thus provides a methodological framework for quantifying COEs at a landscape level, demonstrating that it is feasible to tackle this underexplored area with direct relevance to ecological management and decision-making. Specifically, integrative capture-recapture and path analysis of our nationwide multi-decadal dataset showed that light birds with high haematocrit and buffy coat survived winters much better and can thus be considered in better body condition. Next, we showed that winter body condition carried over to reproductive performance in the subsequent breeding season, leading to higher clutch and hatchling survival. However, the birds did not achieve this by laying earlier. Finally, we found that grassland around wintering areas significantly increased the winter body condition. In addition, temperature and cockle availability have differential impacts on individuals depending on their diet specialisation. While worm specialists benefit from higher winter temperature, shellfish specialist are unaffected by these thermal changes. Conversely, increased cockle availability enhances body condition of shellfish specialists, but does not influence worm specialists. Crucially, neither temperature nor cockle availability trends over recent decades account for

the historic decline in reproductive success. However, our path models revealed that the sharp reduction in grassland area, exceeding 40% in recent decades in the Netherlands, correlates with a reduction in offspring survival of 13.5%, underscoring the pressing need for habitat restoration in wintering grounds. Our study thus supports the hypothesis that causes of reproductive decline throughout the Netherlands are at least partly caused by a COE of a deteriorating winter environment, and that causes do not necessarily occur only in the breeding environment.

Role of body condition in mediating carry-over effects

We used body condition as an individual trait that may mediate the COE. Harrison et al. (2011) emphasized the importance of investigating ‘true’ COEs using individual state variable (e.g. condition, arrival date) to get better insight in the causal mechanism, rather than investigating relations between environmental conditions in one season and the individual performance in the next season. Our study illustrates the added value of including a mediating state variable. A previous study on oystercatchers has shown that good reproductive years follow after cold winters (van de Pol et al. 2010), but the path analysis in our current study shows that this association is not due to a COE, as cold winters tend to lead to reduced body condition. As another example, the result that body condition mediates the effect of grassland on reproduction implies that the influence of grassland is direct rather than merely a proxy for another correlated causal driver in summer. One might speculate that the presence of grasslands in winter inversely correlates with arable land, which could potentially enhance summer reproductive success (e.g. because grasslands, often mowed, pose risks to nests while arable lands less frequently result in machine-induced nest loss). However, if winter grasslands were simply a non-causal proxy linked to arable land, a correlation with winter body condition would not be expected. This suggests a direct role of grassland influencing body condition and subsequent reproductive success.

The positive effect of winter body condition on reproductive success can be explained by a variety of non-mutually exclusive behavioural mechanisms. Individuals that enter the breeding season in better condition may (i) have more energy to chase away nest predators, (ii) need to forage less themselves and thus have more time to guard the nest, and (iii) also have more energy to find food for their chicks. Notably, we did not find a relationship between winter body condition and lay date, suggesting that the advantage of high body condition is not that it allows them to lay earlier (which would be advantageous as early nest produce more offspring; Allen et al., 2022). A possible explanation for the absence of a relationship between condition and lay date may be that the condition variables (body mass, haematocrit and buffy coat) we use to define body condition show relatively strong within-individual variation between different winter seasons. Data from birds being caught twice in different winters reveal that within-individual variation is about 70% for body mass and buffy coat, and 30% for haematocrit (Appendix S1: Table S12). Therefore, body condition is a more dynamic trait, whereas lay date is known to be highly repeatable across years within individuals and relatively constant throughout the lifetime of an individual in our study species (van De Pol and Pettifor 2006).

One of the ways how our study advances on existing COE studies (Bearhop et al. 2005, Gunnarsson et al. 2006, Sorensen et al. 2009) is by taking a holistic view of body condition. By quantifying the body condition based on not only body mass, but also accounting for physiological measures, we expect to have produced a more appropriate multidimensional predictor explaining fitness (Harrison et al. 2011, Frauendorf et al. 2021). Studies focussing solely on body mass (accounted for size) typically consider it as a measure of fat content, but if we had done the same, our results would have been hard to interpret. In our study, size-corrected mass was negatively correlated with our body condition index and overwinter survival, which clashes with the idea that size-corrected mass is a proxy of fat content.

Several possible explanations exist for a negative relationship between mass and condition/survival. First, birds that are very heavy, are more easily caught by predators since they are less agile due to higher mass (Zwarts et al. 1996b). Peregrine falcons (*Falco peregrinus*), potential predators of oystercatchers in their wintering ground, have been increasing nationwide (Appendix S1: Figure S16), which may have resulted in stronger selection on being lean. Second, birds in better condition may also be more dominant, which

allows for a more constant access to food sources and diminishes the need to store fat reserves in preparation for worse times (compared to subdominants; Cuthill et al., 2000; Ekman & Liliendahl, 1993). Third, if the underlying relationship between mass and condition/survival is parabolic, we may not have sampled enough individuals that were close to starvation (low body mass and low survival) and therefore only detected a negative linear relationship between mass and condition. This could be possible, as our catching years all involved warm benign winters, and we found no model support for a non-linear relationship; Appendix S1: Table S6).

Our reported positive relationship between haematocrit and condition/survival echoes previous studies on our study species (Verhulst et al. 2004), bar-tailed godwits (Piersma et al. (1996) and crimson fitches (*Neochmia phaeton*) (Milenkaya et al. 2015). Our observed negative relationship between buffy coat and condition/survival is consistent with the idea that high levels of white blood indicate a well-working immune system (Hanssen et al. 2005). However, a previous study on our study species found a positive relationship between buffy coat and body condition Verhulst et al. (2004), which is consistent with the opposing idea that the fraction of white blood cells becomes elevated if the body fight against infections (Ullman-Culler 1999, Campbell et al. 2008, Beechler et al. 2012). We note that Verhulst et al. (2004) operationalized body condition using different methods based on the pattern of correlation among condition variables using principal component analysis, and not by how well it explained variation in survival. In fact, in the same paper Verhulst et al. (2004) reported that oystercatchers with low buffy coat were more likely to be reported dead, suggesting that consistent with our study, also in their study buffy coat was positively related to subsequent survival. For a more in-depth discussion of the differences between ours and the Verhulst et al. (2004) study, see Appendix S1: Text S4.

Despite the importance of investigating ‘true’ COEs to understand population dynamics, these kinds of studies also pose serious challenges. Following individuals throughout their annual life cycle and collecting data year-round (or at least during two seasons) such that analyses are powerful enough to detect effects, is a major challenge (Harrison et al. 2011). In our study, finding individuals back during the breeding season was very hard. Despite a huge amount of ring reading effort by citizen scientists (typically >100 daily resightings across the country during the breeding season), we could only locate the breeding territory of 4.3% of birds caught in the previous winter. Equipping birds with GPS-tracking would have increased the sample size (for the carry-over effect analysis; Figure 1e), however, this also requires larger budgets and GPS-trackers can have undesirable effects on body condition. Despite being able to only follow few individuals across seasons, we found clear evidence for COEs, as effect sizes were strong. However, strong COEs may not be general in other studies, and additional less strong COE pathways may have gone undetected in our study.

Winter environmental effects carrying over on reproduction

We found that the available grassland around a catching site had a strong positive effect on winter body condition. Previous studies suggested that oystercatchers foraging on grasslands in winter could be seen as indication that individuals have problems finding enough food during low tide on the intertidal flats (Heppleston 1971, Caldow et al. 1999, Durell et al. 2001). Especially worm specialists make use of it as an additional food source during high tide (when intertidal flats are not exposed for foraging) but also during low tide as an alternative food source (van der Kolk et al. 2019). Our results suggest that grasslands are not necessarily marginal foraging habitat and are more important for wintering oystercatchers than thus far appreciated

Winter temperature had a positive effect on the winter body condition, but only for worm specialists. Worm specialists depend more on grassland foraging, but during cold spells they cannot use grasslands as invertebrates are unreachable in the frozen soil. In addition, soft-bodied prey like ragworms (*Nereididae*), which are an important food source in intertidal flats for worm specialists (Van de Pol et al., 2009), are more difficult to obtain under colder conditions than shellfish as they become less active.

Higher cockle availability also was associated with an increase in body condition, but only in shellfish specialists. However, this effect is not as strong as from the other environmental variables (Figure 5), suggesting

that the body condition, how we defined it, is not strongly affected by cockle availability as food source (nor by mussels).

Our reported differences among diet specialists in how their body condition responds to environmental variability add to previous studies showing that worm and shellfish specialists differ in their survival and environmental sensitivity of foraging time (Van der Kolk et al., 2019). Individual variation in responses to environmental variability is of general relevance for COE studies. COEs at the individual level do not necessarily impact the population level, as opposing effects on different types of individuals can cancel each other out. For example, we found that the environmental impact varied among diet specialists, but at the population level, the environmental dependency of body condition is less clear because the individuals benefiting from an environmental change are counterbalanced by other individuals suffering from the same environmental change. As we have information on how the proportion of diet specialists changed over time (Appendix S1: Figure S17), we can delve deeper into the effect on the population level. The proportion of worm and shellfish specialists has decreased by 8% and 6%, respectively, from the early 2000s (Appendix S1: Table S13). In contrast, the proportion of individuals with an intermediate bill shape increased by 14% (Appendix S1: Table S13). This suggests that the population as a whole has become less susceptible to temperature and shellfish fluctuations in the past two decades.

Deteriorating winter environmental conditions and ongoing reproductive and population decline

Based on the migratory connectivity (shared overwintering habitat, many different breeding habitats) of oystercatchers, we hypothesized that there may be shared environmental drivers in the wintering ground that all individuals experience which may explain the low reproductive success across the Netherlands since the 1990s. The only variable that received strong statistical support and that could support this hypothesis would be the grassland proportion around the wintering ground. The size of permanent grasslands decreased by almost 40% across the Netherlands since 1980s, which we showed could account for a large (13.5%) reduction in hatchling survival over the same period and could thus be an important and thus far ignored driver of ongoing population decline of Dutch oystercatchers.

Therefore, we recommend preserving and restoring grasslands, especially permanent grasslands (rather than changing them to arable land or even buildings) close to highly used high tide roost areas to ensure that the birds can forage on grasslands close to their roost areas. Permanent grasslands are more valuable than temporary grasslands or arable land, as their lower soil disturbance frequency results in higher earthworm abundance (Onrust et al. 2019), the main food source of grasslands-feeding oystercatchers (Zwarts et al. 1996b). Ultimately, the findings of our study (e.g. effect sizes of pathways) can be included in population models to compare the relative importance of COEs of environmental change in winter on the reproductive success compared to environmental drivers acting directly in the breeding ground, to better understand the relative importance of winter and summer drivers on the demography of the Dutch oystercatcher population.

Conflict of interest statement

All authors declare no conflict of interest.

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Figure captions

Figure 1: Conceptual framework and statistical models (depicted as path diagrams) used to analyse carry-over effects. a) Conceptual diagram describing how environmental variables affect the state variable body condition that can mediate carry-over effects of winter environment on breeding success in the summer. Overwinter survival is used to identify which body condition variables can usefully operationalize and quantify the latent concept body condition (while correcting for confounding effects of such as handling time and individual characteristics such as age and sex). The relationship between these variables was analysed using a statistical for the winter part (multi-state mark-recapture structural equation model; multi-state CR-SEM; dotted line; detailed in panels b, c & d) and a structural equation model for the summer part (SEM; dashed line, panel e). Also shown are sub-models for body condition (b), for estimating survival (and how it depends on body condition) for capture mark recapture data (c), for determining the impact of winter environmental variables on body condition (d), and for quantifying direct and indirect (via early laying) impacts of winter body condition on breeding success in summer. Directional arrows refer to the direction of the hypothesized causal relationship. Note that the capture site effect in (d) is also included in the survival analysis (c) by having a ‘state’ (henceforth called site) and time effect.

Figure 2: Map of the catching locations across the Wadden Sea (a) and the Delta area (b) as well as the area-division (geographical sites; tidal basins) used in the multi-site mark-recapture model and for the nationwide citizen science resightings. (c) shows an overview map of the locality of the two overwintering estuaries within the Netherlands. Dots and diamonds indicate catching events in the winter of period 1 (2000-2003) and period 2 (2016-2018), respectively. The size of the symbols indicates the number of caught birds per catching event. Stars indicate the weather station used in the analysis, with the nearest weather station to each catching location indicated by the same colour (Appendix S1: Table S4). Orange, blue, grey and green star indicate the weather stations in De Kooy, Hoorn Terschelling, Lauwersoog and Vlissingen, respectively.

Figure 3: The effect sizes in the model for body condition. (a-c) The effect of confounding variables (e.g. age, sex, handling time) on the three condition variables. (d-f) The effect of the size-corrected mass, haematocrit and buffy coat on body condition, respectively. (g) The relationship between body condition and survival probability. In (a-c) circles and error bars indicate the median and 95% credible interval, respectively. Note that for the categorical variables (age and sex), it is the difference between the classes that is of interest, rather than whether it overlaps with an effect size of zero. In (d-f) the predictor variables (condition measures) have been z-score standardized (mean=0, sd=1) before the analysis and the values shown here are standardized as well as corrected for the confounding variables (note that the intercept for composite variable condition is 0, resulting in no credible interval at point (0,0)). In (g) the fitted line is the survival probability for an average site and year. The dots represent the predicted survival probability from the multi-state model, which is why the points from a site/ year/age combination are aligned along logistic curves. In addition, (g) is a contrast plot meaning that the reference group of the survival probability for the site/age combination is 0 and the other levels are plotted in contrast to the 0 (thus the credible interval is 0 at condition 0). For the effect of body condition on survival per age class, see Appendix S1: Figure S14.

Figure 4: Results from the separate path analyses of (a) body condition on survival or (b) lay date as well as (c) lay date on survival. Clutch survival is represented in orange ($n=32$) while hatchling survival is represented in blue ($n=25$). Also given are the posterior mean, 95% credible interval in brackets and probability of direction (p_d) along the path ways. The fitted and binned data points (averaged per levels of the independent variable) are plotted because of the strong effect of crop fields on lay date and hatchling survival (Appendix S1: Figure S15). Symbol size is proportional to sample size. We visualized the fitted line (solid for an at least moderately strong effect and dashed for no evidence of an effect) and the shaded area represents the 95% credible interval.

Figure 5: The effect of environmental variables (weather (a-b), density (c) and food condition (d-f)) on winter body condition. Data are aggregated per sample year and site. Lines depict the fitted regression model, with shaded areas showing the 95% credible interval. Values in the left upper corner show the posterior mean of the regression coefficient with the 95% credible interval in brackets as well as the probability of direction p_d . Solid and dashed line indicate evidence or no evidence of a main effect of the environmental variable

on body condition (based on p_d), respectively. Note that temperature and precipitation are latent variables informed by various temperature and precipitation metrics (see Methods).

Figure 6: (a) Standardized parameter estimates (posterior median $\pm 95\%$ CI) of the effect of diet specialization on (a) winter body condition and (b) the effect (posterior median $\pm 95\%$ CI) of environmental variables on condition per diet specialization (interaction effect). For an overview of the distribution of diet specialization across the study population see Appendix S1: Table S10.

Figure 7: Overview of the main results of the study. Signs (-/+) indicate the direction of the relationship. The row ‘change over time’ indicates in which direction the variable changed over the last decades. The arrows in the left box of ‘total effect’ indicate an increase in the variable and the total effect of this increase through body condition on the hatchling survival. For instance, a grassland proportion increase of 0.13 in winter increases the hatchling survival by 4.5%, which is equal to 0.8 extra days of the survival of the hatchling. The right panel of ‘total effect’ box shows the effect of the decline of grassland proportion by 40% since the 1980s on the hatchling success through body condition. Note that we only show here the three environmental variables that had at least a moderate strong effect (p_d) in the main and interaction model.

Figures

Figure 1

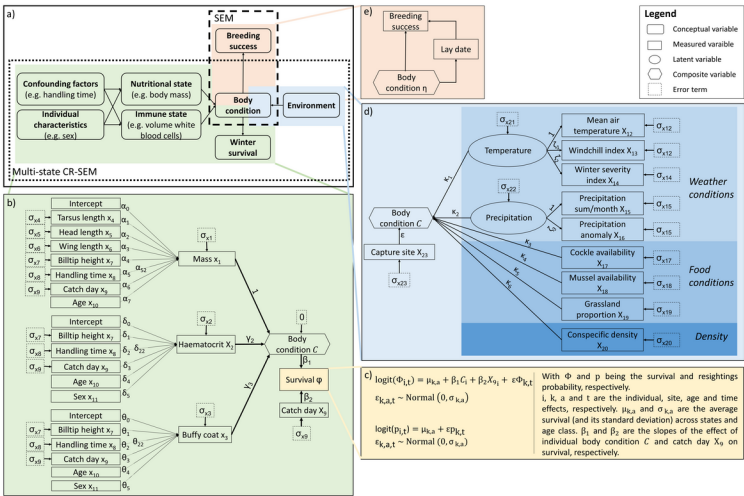


Figure 2

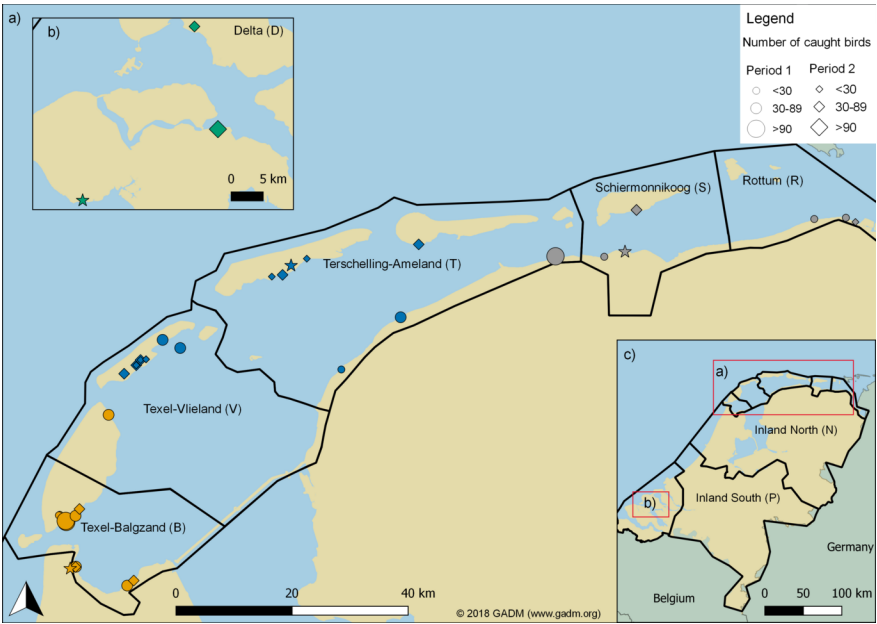


Figure 3

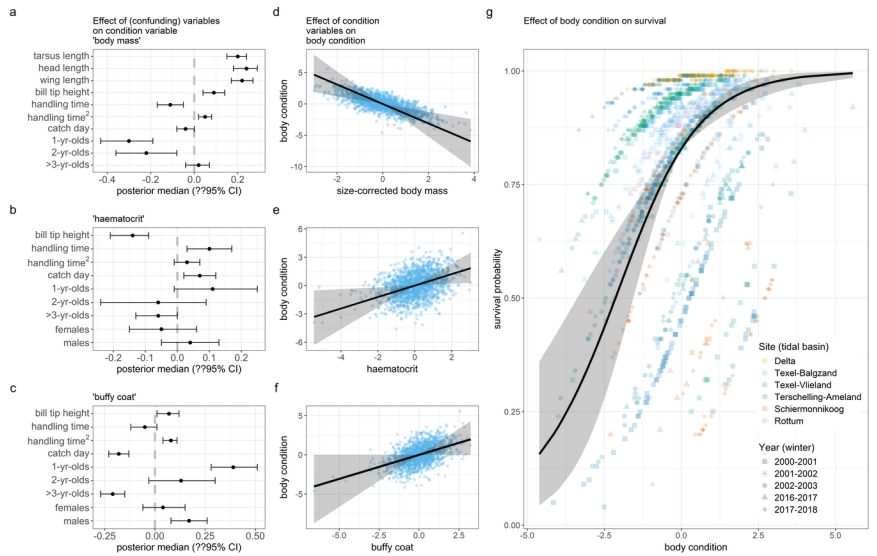


Figure 4

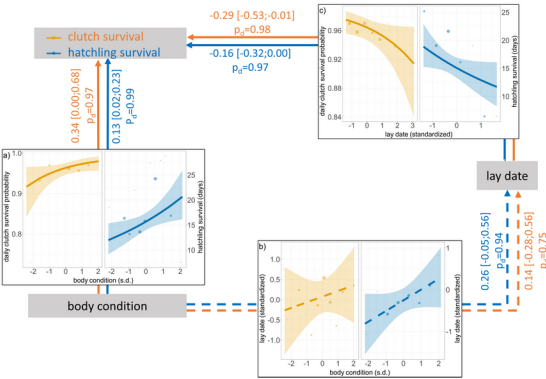


Figure 5

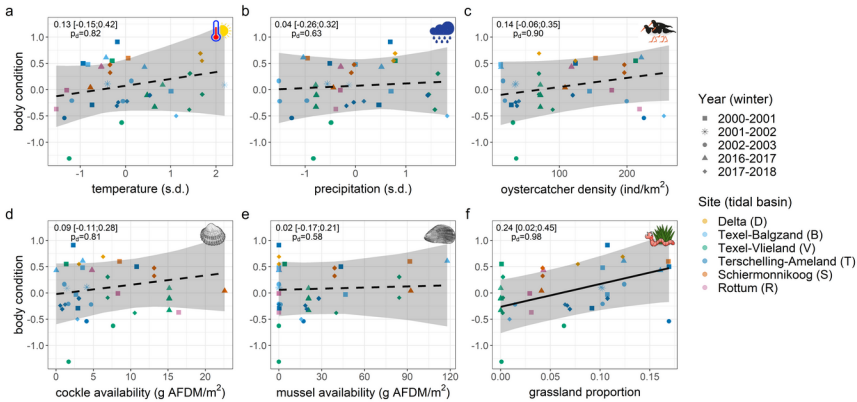


Figure 6

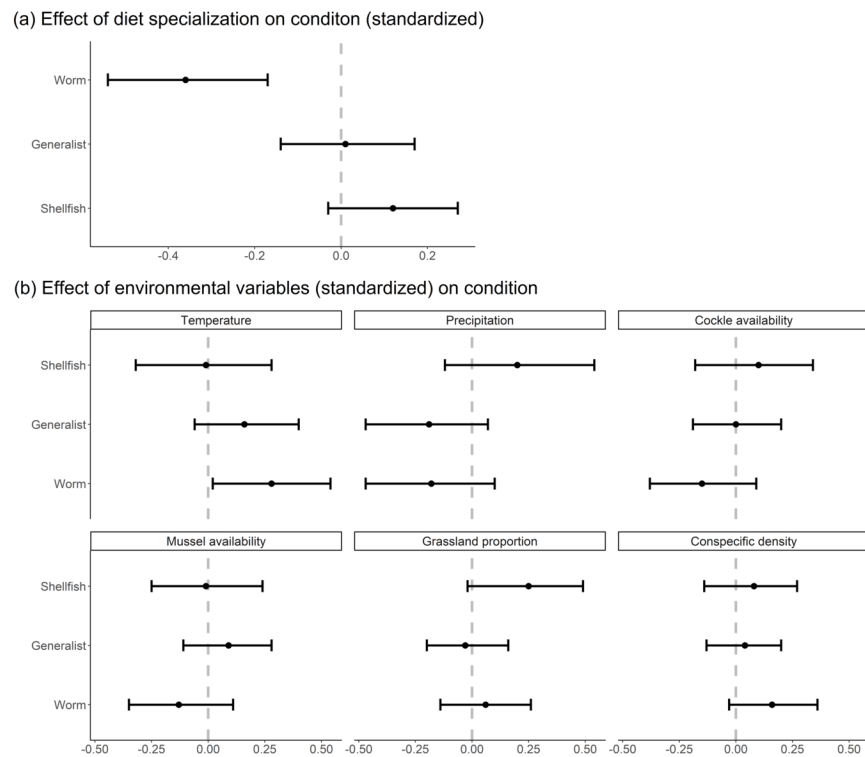
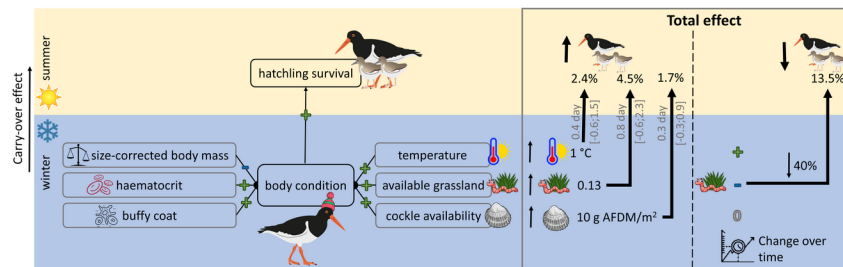


Figure 7



Supporting Information

Appendix S1: Supplemental tables, figure and text

Appendix S2: Technical details of the multi-state mark-recapture model

Appendix S3: Density and trace plots of the multi-state CR-SEM

Appendix S4: Potential scale reduction factor (PSRF, R-hat) of the multi-state CR-SEM

Appendix S5: Parameter estimates of the variables from the multi-state CR-SEM